(28) $\dfrac{4\sqrt{x}\sqrt{x+\sqrt{x}}+2\sqrt{x}+1}{8\sqrt{x}\sqrt{x+\sqrt{x}}\sqrt{x+\sqrt{x+\sqrt{x}}}}$ (29) $\dfrac{\cos\left(\tan\sqrt{\sin x}\right)\sec^2\left(\sqrt{\sin x}\right)\cos x}{2\sqrt{\sin x}}$ (30) $\dfrac{-2}{1+x^2}$

Exercise 10.4

(1) $x^{\cos x}\left(\dfrac{\cos x}{x}-\sin x\log x\right)$ (2) $x^{\log x}\left(\dfrac{2\log x}{x}\right)+(\log x)^x\left[\dfrac{1}{\log x}+\log(\log x)\right]$

(3) $\dfrac{y(2x-1)}{x(1+2y)}$ (4) $\dfrac{y(x\log y-y)}{x(y\log x-x)}$ (5) $(\cos x)^{\log x}\left[\dfrac{\log(\cos x)}{x}-\tan x\log x\right]$

(6) $-\dfrac{b^2x}{a^2y}$ (7) $\dfrac{x\sqrt{x^2+y^2}+y}{x-y\sqrt{x^2+y^2}}$ (8) $\dfrac{1-\sec^2(x+y)-\sec^2(x-y)}{\sec^2(x+y)-\sec^2(x-y)}$

(10) $\dfrac{1}{2}$ (11) $\dfrac{6}{1+9x^2}$ (12) 1 (13) $-\tan t$

(14) $\tan t$ (15) $\dfrac{t^2-1}{2t}$ (16) $\dfrac{2}{1+x^2}$ (17) $\dfrac{3}{\sqrt{1-x^2}}$

(18) 1 (19) $\cos x^2$ (20) 2 (21) $\dfrac{1}{2}$

(22) -1 (23) $\dfrac{x}{(1-x^2)^{\frac{3}{2}}}$

Exercise 10.5

(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)
2	4	3	3	1	4	3	2	1	4	3	3	2	2	4
(16)	(17)	(18)	(19)	(20)	(21)	(22)	(23)	(24)	(25)					
2	3	1	4	2	4	1	1	3	2					

Exercise 11.1

(1) (i) $\dfrac{x^{12}}{12}+c$ (ii) $-\dfrac{1}{6x^6}+c$ (iii) $\dfrac{3}{7}x^{\frac{7}{3}}+c$ (iv) $\dfrac{8}{13}x^{\frac{13}{8}}+c$

(2) (i) $-\cot x+c$ (ii) $\sec x+c$ (iii) $-\text{cosec}x+c$ (iv) $\tan x+c$

(3) (i) 12^3x+c (ii) $\log|x|+c$ (iii) e^x+c

(4) (i) $\tan^{-1}x+c$ (ii) $\sin^{-1}x+c$

Exercise 11.2

(1) (i) $\dfrac{(x+5)^7}{7}+c$ (ii) $\dfrac{1}{9(2-3x)^3}+c$ (iii) $\dfrac{2}{9}(3x+2)^{\frac{3}{2}}+c$

(2) (i) $\dfrac{-\cos 3x}{3}+c$ (ii) $-\dfrac{\sin(5-11x)+c}{11}$ (iii) $-\dfrac{\cot(5x-7)}{5}+c$

(3) (i) $\dfrac{1}{3}e^{3x-6}+c$ (ii) $-\dfrac{e^{8-7x}}{7}+c$ (iii) $-\dfrac{1}{4}\log|6-4x|+c$

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